Air Quality, Cost of Living, and Crime Across U.S. Counties

Two-page summary of a capstone project · Revised edition, October 2026

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Bottom line. Across 2,892 U.S. counties, crime is most strongly associated with *lower income* and *larger, more urban populations*. Cost of living and air quality, the two features this project set out to test, are not reliably associated with crime once the data are analyzed at the right level. A first version of this project reported both as "highly significant" ($p < 10^{-16}$); each result turns out to be an artifact, for a different reason.

Question and Motivation

People often compare places by crime, cost of living, and environmental quality when deciding where to live. Research links crime to density and disadvantage [1, 2] and, more recently, to air pollution and lead exposure [3, 4]. I ask: (1) Is air quality associated with county crime rates? (2) Is cost of living? (3) Which available features matter most when considered together?

Data

A public quality-of-life dataset on Kaggle [5], compiled from U.S. government sources: reported crimes per 1,000 residents in 2016 (FBI Uniform Crime Reporting), the share of days with a "good" Air Quality Index (EPA), family cost of living and median income (Economic Policy Institute), unemployment (USDA), 2022 population (Census), drinking-water violations (EPA), and NCES school locale codes, from which I build the share of a county's schools in cities or suburbs. **Key fact:** air quality is recorded *once per state*, so it has only 50 distinct values regardless of how many rows are analyzed. After dropping counties reporting zero crime (likely non-reporting) and Illinois (absent from the school file), the sample has 2,892 counties in 49 states and D.C.; mean crime is 19.8 per 1,000.

Method

OLS regressions of county crime on air quality, cost of living, log population, urban share of schools, unemployment, log median income, and the cost-to-income ratio. Standard errors are clustered by state (50 clusters, t_{49} inference). Further models add state, Census region, or Census division fixed effects; I also estimate at the state level ($n = 50$) and report standardized coefficients to compare variables on a common scale.

Findings

1. **The original significance came from a fake sample size.** The first version regressed crime on air quality using 87,457 school rows that repeat county and state values. Clustering by state makes the standard error 15 times larger; t falls from -67.4 to -4.5 for the same slope.
2. **Cost of living has no association with crime** (slope -0.33 per \$10,000, 95% CI $[-1.64, 0.98]$, $p = 0.62$). Expensive counties are richer (less crime) but also more urban (more crime), and the two cancel. Holding income and urbanization fixed, counties that are expensive *relative to income* have somewhat less crime ($p = 0.008$).
3. **Air quality mostly stands in for geography.** Alone, one SD more good-air days goes with 10% less crime ($p < 0.001$). Controls halve this (-0.15 per point, $p = 0.028$). Comparing counties within the same Census region cuts it by another 60% (-0.06, $p = 0.36$; divisions: $p = 0.49$), although region explains only 13% of the variation in air quality and the standard error is unchanged. The states with the fewest good-air days are in the West and Southwest (Figure 1).
4. **Income and urbanization dominate** (Figure 2). One SD higher median income goes with 0.55–0.64 SD less crime (about 2.6 fewer crimes per 1,000 for 10% higher income); population size (+0.38–0.46 SD) and urban share (+0.26–0.28 SD) follow. Unemployment and drinking-water violations show no association. Controls raise R^2 from 0.03 to 0.32; state fixed effects raise it to 0.45.

Robustness and Limitations

The step-2 estimate is stable to keeping zero-crime counties, adding water violations or the state vote winner, swapping cost-to-income for cost of living, or restoring Illinois ($p = 0.02$–0.05), but not to population weighting ($p = 0.19$) or regional comparisons. Limitations: a single cross-section (associations only); air quality measured only by state; **reversed timing**, since crime is from 2016 while income, unemployment, and population are from 2022; total reported crime only, rounded and not split into violent and property crime; no land area, poverty, or age structure.

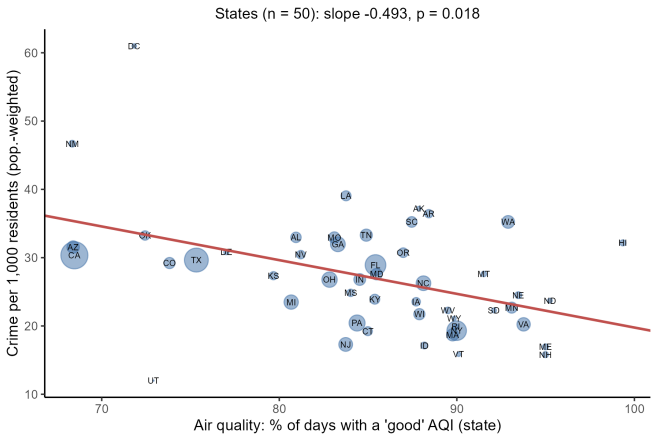


Figure 1: Crime vs. air quality across states (population-weighted; point size = population). Low good-air states cluster in the West.

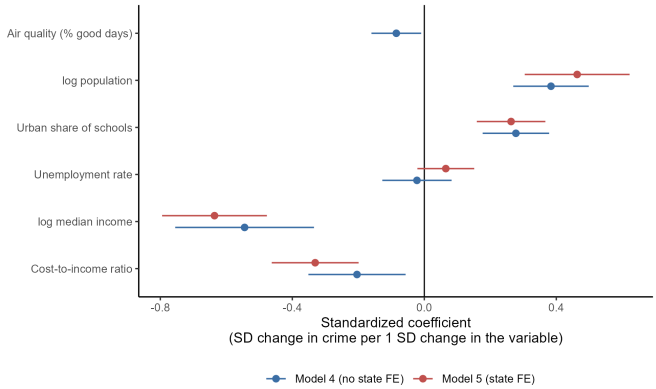


Figure 2: Standardized coefficients with 95% CIs; Model 5 compares counties within a state (air quality drops out).

Table 1: The air-quality slope (crimes per 1,000 per percentage point of good-air days) as the analysis is corrected step by step.

Step	Specification	Slope	SE	p
0	School rows, conventional SE (first version)	−0.369	0.005	$< 10^{-16}$
1	Counties, SE clustered by state	−0.303	0.085	< 0.001
2	+ population, urbanization, income, unemployment, cost-to-income	−0.154	0.068	0.028
3	+ Census region fixed effects	−0.063	0.069	0.365
4	+ Census division fixed effects (instead of regions)	−0.051	0.072	0.488

Conclusions

Income and urbanization, not cost of living or state-level air quality, are the features most strongly associated with county crime. The result does not contradict studies finding that short-run pollution spikes raise violent crime [3]; a state average is too coarse to detect such effects. Methodologically, the project shows three common ways a "significant" result can be spurious: repeating group-level values across rows [6, 7], confounders that offset each other, and an area-level variable that proxies for geography. A county-year panel with satellite $PM_{2.5}$ and violent-crime data would be the natural next step.

References

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